# B3.2 Transport

| NAME | DATE | TOTAL |
|---|---|---|
| | | _____ / 53 |

**About 50 minutes,** or one section at a time. A calculator is allowed. The number of marks shows how many separate points to make. Look at the bold command term: "state" needs a brief answer, while "explain" needs reasons.

## Section A: Recall

9 MARKS

1 STATE two features of capillaries that make exchange efficient. [B3.2.1] [2]

2 STATE the function of valves in veins. [B3.2.5] [1]

3 IDENTIFY the artery felt when a pulse is taken at the wrist. [B3.2.4] [1]

4 STATE what is meant by occlusion of an artery. [B3.2.6] [1]

5 STATE the tissue in an artery wall that stretches and then recoils. [B3.2.3] [1]

6 STATE why no pulse can be felt in a vein. [B3.2.5] [1]

7 STATE two possible consequences of a blocked coronary artery. [B3.2.6] [2]

## Section B: Understanding

15 MARKS

8 EXPLAIN how capillaries are adapted for the exchange of materials with tissues. [B3.2.1] [4]

9 **EXPLAIN** how the wall of an artery helps it to withstand and maintain high blood pressure. [B3.2.3] [3]

10 **EXPLAIN** how blood returns to the heart in veins, despite the low pressure. [B3.2.5] [3]

11 **COMPARE AND CONTRAST** the structure of arteries and veins. [B3.2.2] [3]

| ARTERIES | VEINS |
|---|---|

12 **SUGGEST** why capillaries in the small intestine have fenestrations, while those in most muscle do not. [2]
[B3.2.1]

## Section C: Working with data

20 MARKS

*All data in this section are invented for practice. The patterns are realistic, but they do not describe real people or countries.*

13 [B3.2.2]

A student measured two blood vessels on a micrograph of a cross-section.

| VESSEL | WALL THICKNESS / MM | LUMEN DIAMETER / MM | WALL : LUMEN RATIO |
|---|---|---|---|
| X | 1.2 | 4.0 | |
| Y | 0.4 | 5.0 | |

a **CALCULATE** the two missing ratios of wall thickness to lumen diameter. [2]

b **DEDUCE** which vessel is the artery, giving a reason. [2]

14 [B3.2.4]

A student counted their resting pulse for 30 s on three occasions: 36, 38 and 37 beats. They then exercised for three minutes and measured their heart rate each minute during recovery.

| MINUTES AFTER EXERCISE | 0 | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|---|
| HEART RATE / BEATS PER MINUTE | 140 | 118 | 100 | 88 | 78 | 74 |

a CALCULATE the mean resting heart rate in beats per minute. [2]

b CALCULATE the percentage change in heart rate from the resting value to 0 minutes after exercise. [2]

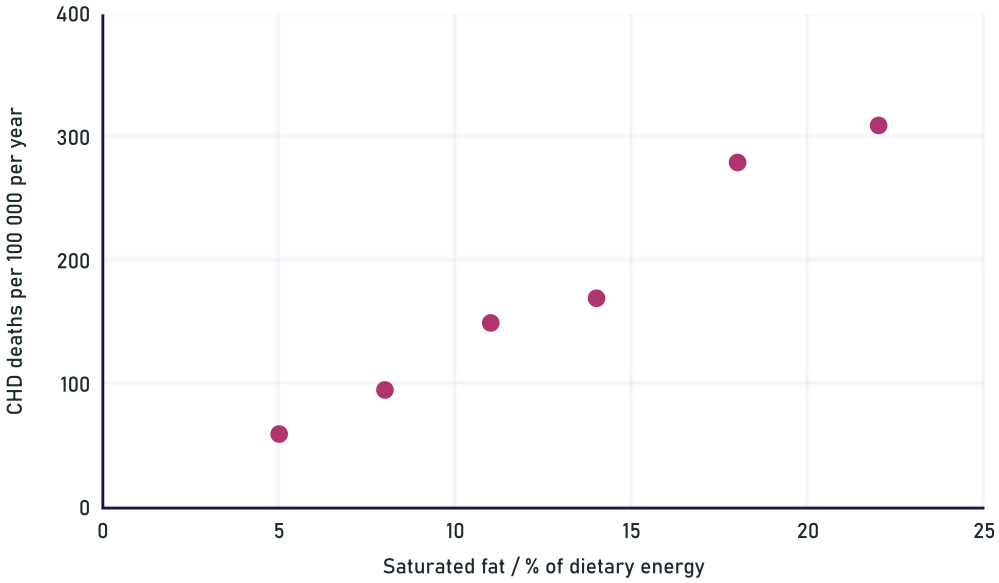
c DESCRIBE the pattern of recovery. [2]

d EXPLAIN why the pulse is taken with the fingertips, at a point where an artery passes over bone. [2]

e COMPARE this method with using a fitness watch, giving one advantage of each. [2]

15 [B3.2.6]

The graph shows mean saturated fat intake and deaths from coronary heart disease (CHD) in six fictional countries.




a DESCRIBE the relationship shown. [1]

b EVALUATE the claim that these data prove saturated fat causes CHD. [3]

c SUGGEST two other variables that could differ between these countries and affect CHD deaths. [2]

## Section D: Longer answers

9 MARKS

16 EXPLAIN what makes coronary arteries become blocked, and what follows. [B3.2.6] [6]

17 **SUGGEST** why a thrombus in a coronary artery can damage only part of the heart, and why that can still be fatal. [B3.2.6] [3]